

**Time:** 5 minutes**Outcome:** Analyze Short-Run Production & Diminishing Marginal Product.**Difficulty:**

Intermediate

Short-Run Production & Diminishing Marginal Product



THE CORE IDEA

In the short run, at least one input is fixed, so output changes as additional units of a variable input are employed. Input flexibility is the production-side basis for distinguishing fixed and variable cost. The same arithmetic relationship governs MP versus AP and MC versus AVC or ATC.



HOW TO RECOGNIZE IT

- Production case 2: Ironwood Furniture has MP sequence 6, 10, 13, 12, 10, 8, 6, 4.
- Production case 4: Cedar Trail Bikes has MP sequence 9, 12, 13, 11, 9, 6, 2, -2.
- Production case 6: Riverbend Textiles has MP sequence 11, 14, 15, 13, 11, 9, 7, 5.
- Production case 8: Summit Glassworks has MP sequence 12, 15, 16, 14, 12, 10, 8, 6.

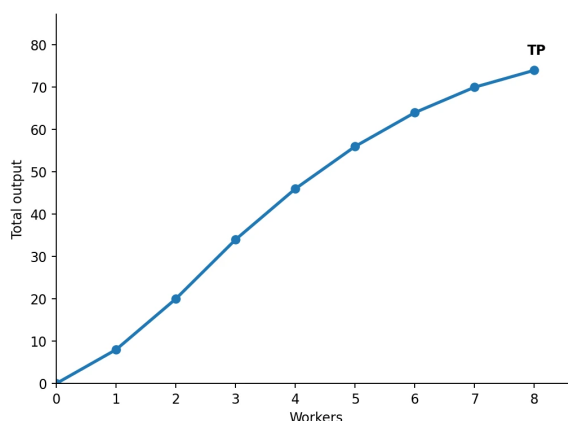


WATCH OUT

A common mistake is to wait until mp negative. Identify the first decline in MP and separate it from negative MP and long-run scale.



WORKED EXAMPLE: DIMINISHING MARGINAL PRODUCT



Total product rises from 34 units with three workers to 46 with four, so the fourth worker's marginal product is 12. The fifth worker raises output from 46 to 56, giving marginal product of 10. Output still increases, but the smaller addition shows diminishing marginal product.



CHECK YOURSELF

In the short run, which statement must be true?

**READY?**

Return to the game and master this concept.